

On the headedness of Dutch quantitative ER. Ellipsis interactions and long head movement

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Abstract

One of the functions of the Dutch pronominal element ER is to resume an elided quantified noun. This paper argues that this quantitatively used ER is a head that undergoes long head movement. Evidence comes from the interaction of quantitative ER with gapping and NP ellipsis. The first observation is that ER is obligatorily elided under gapping. It is argued that ER moves to a head position, which forces its deletion under gapping, because only A-bar specifiers can be remnants of gapping (Broekhuis, 2018). It is then argued, contra Barbiers (2017); Kranendonk (2010), that the domain of ER does not include adjectives and PP complements, based on cooccurrence restrictions of ER and adjectives in Dutch dialects, and the distribution of complements in other NP ellipsis contexts; rather, ER is a head. It is argued that ER can undergo long head movement because it is a minimal and maximal projection at the same time. This accounts for its mixed head and A-bar movement properties: the structural size and landing site of ER is that of a head, but it can move long-distance and license parasitic gaps. As such, this paper is a detailed investigation of the properties of long head movement.

Keywords: Dutch, quantitative ER, gapping, NP ellipsis, long head movement

1 Introduction

The Dutch pronominal element ER has multiple uses, which are illustrated in (1). Linguists have long been fascinated by the various uses of ER because in each of its uses, ER has slightly different properties and distribution.

- (1) a. Er staat een man op tafel. (expletive ER)
ER stands a man on table
'There is a man standing on the table.'

- b. Hij staat er op. (prepositional ER)
he stands ER on
'He stands on it.'
- c. Hij staat er graag. (locative ER)
he stands ER happily
'He likes to stand there.'
- d. Hij verft er twee. (quantitative ER)
he paints ER two
'He's painting two.'

This paper aims to gain insight into the properties of ER, with a focus on quantitative ER (Q-ER), by looking at its syntactic structure. Many previous studies on Q-ER have argued that Q-ER is a phrase (Barbiers, 2017; Bennis, 1986; Blom, 1975-1976; Coppen, 1985; Kranendonk, 2010; van Riemsdijk, 1978).¹ This paper argues, based on the interaction between Q-ER and various types of ellipsis, that Q-ER is a head that undergoes long head movement. I show that this proposal can account for the distribution and movement properties of Q-ER, which show a mix of head and A-bar behaviour. As such, this paper is a detailed exploration of what long head movement looks like.

The paper is structured as follows. In section 2, I explore the observation that ER obligatorily deletes under gapping. I apply movement diagnostics to clauses with gapping to demonstrate that ER is gapping from its derived position. ER deletion under gapping follows from Broekhuis (2018)'s analysis of gapping, according to which gapping elides everything but A-bar positions: the derived position of ER is shown not to be an A-bar position, but rather, a head position. Section 3 argues that Q-ER is a head. The first argument comes from the dialectal distribution of the cooccurrence of Q-ER and adjectives. It is argued that there is a correlation between licensing strategy of NP ellipsis with adjectival remnants, and cooccurrence of Q-ER and adjectives: Q-ER only cooccurs with adjectives in those dialects that do not pronominalise N in the contexts of adjectival remnants of NP ellipsis. This follows if Q-ER is also a pronominalisation of N. I also argue that Q-ER cannot take complements because it is unable to select them, not because complements are included in the domain of Q-ER, again pointing to Q-ER's status as a N head. Section 4 considers how ER moves. I argue that Q-ER has a mix of A-bar and head properties because Q-ER pronominalises a N head that is minimal and maximal at the same time. This enables long head movement of Q-ER. Section 5 concludes with a brief comparison of Q-ER and other types of ER, and French *en*.

2 Q-er deletion under gapping

In the context of an indefinite quantificational remnant of NP ellipsis, Q-ER is generally obligatory (2-a). The only exception is when the finite verb undergoes gapping: in that case, Q-ER needs to be absent (2-b).

¹An exception is the unpublished handout by Corver, van Koppen, and Kranendonk (2009), in which they argue that Q-ER spells out *n* when *n*'s complement is elided.

- (2) a. Ik heb twee boeken gelezen, en jij hebt *(er) vijf geschreven.
 I have two books read and you have ER five written
 b. Ik heb twee boeken gelezen, en jij ~~hebt~~ (*er) vijf geschreven.
 I have two books read and you have ER five written
 ‘I have read two books, and you have written five.’

This section shows that Q-ER is present in examples like (2-b), and undergoes movement, but that it is elided as the result of gapping. This follows from [Broekhuis \(2018\)](#)’s analysis of gapping in Dutch, according to which gapping elides everything that is not in an A-bar position. I argue that Q-ER cannot be in an A-bar position, from which it follows that Q-ER is obligatorily deleted under gapping. Instead, Q-ER moves to a head position.

2.1 Q-er movement in sentences without gapping

Previous research on Q-ER has established that ER originates inside the NP and subsequently moves to its surface position ([Barbiers, 2017](#); [Coppen, 1985](#); [Kranendonk, 2010](#)). This means that the underlying structure of (2-a) is as in (3).

- (3) Ik heb twee boeken gelezen, en jij hebt er vijf ~~er~~ geschreven.
 I have two books read and you have ER five ER written
 ‘I have read two books, and you have written five.’

The first argument for ER movement is based on the argument-adjunct distinction concerning movement. Movement is generally possible from arguments, but excluded from adjuncts, because adjuncts are islands for extraction. In example (4-a), the NP *twee dagen* is an argument of the verb *doorbrengen* (‘to spend’). Q-ER pronominalisation of the noun *dagen* is allowed (4-b).

- (4) a. Ik heb twee dagen in Spanje doorgebracht.
 I have two days in Spain spent
 b. Ik heb er twee in Spanje doorgebracht.
 I have ER two in Spain spent
 ‘I’ve spent two days in Spain.’ ([Barbiers, 2017](#), 167)

In contrast, the NP *twee dagen* in (5-a) is an adjunct: it is optional, and the verb *blijven* ‘to stay’ is an intransitive unaccusative as shown by the use a form of ‘to be’ as the auxiliary. In this case, Q-ER pronominalisation of the noun *dagen* leads to ungrammaticality (5-b). The contrast between (4-b) and (5-b) follows automatically if these sentences involve movement of ER from the NP.

- (5) a. Ik ben twee dagen in Spanje gebleven.
 I am two days in Spain stayed
 b. *Ik ben er twee in Spanje gebleven.
 I am ER two in Spain stayed.
 ‘I stayed in Spain for two days.’ ([Barbiers, 2017](#), 167)

The second argument that ER moves comes from parasitic gap licensing. Parasitic gaps need to be licensed by movement, and ER pronominalisation is able to license a parasitic gap. Consider the sentences in (6). In example (6-a), the parasitic gap is not licensed because no movement takes place. In (6-b), where there is Q-ER pronominalisation, the presence of the parasitic gap is fine. Since the only difference is pronominalisation of the noun by ER, ER must be the element that moves.

- (6) a. *Ik heb je [zonder *pg* in te kijken] één boek toegestuurd.
 I have you without in to look the book sent
 b. Ik heb je er [zonder *pg* in te kijken] één toegestuurd.
 I have you ER without in to look one sent
 ‘I have sent you the book/one without looking in it.’ (adapted from Barbiers, 2017, 168)

The final argument for Q-ER movement is that Q-ER pronominalisation of the noun of a single conjunct in a coordination violates the Coordinate Structure Constraint. (7-a) illustrates that Q-ER can pronominalise both nouns of a coordination. However, if Q-ER pronominalises only one of the nouns, this leads to ungrammaticality (7-b).

- (7) Over boeken gesproken... ‘*Speaking about books...*’
 a. Ik heb er vorige maand twee gelezen en drie verkocht.
 I have ER last month two read and three sold
 ‘I read two and sold three last month.’
 b. *Ik heb er vorige maand twee en drie tijdschriften gelezen.
 I have ER last month two and three magazines read
 ‘I read two (books) and three magazines last month.’

The three tests show that Q-ER moves from an NP-internal base position to an NP-external derived position. This means that in principle, Q-ER could be deleted under gapping in either position. If Q-ER is deleted from its base position, sentences with gapping would not show evidence for Q-ER movement. ER deletion under gapping could then be treated as parasitic ellipsis, i.e. ellipsis from remnants of gapping (Fitzgibbons, 2014; Schneider, 2023). If Q-ER is deleted in its derived position, sentences with quantificational remnants of NP ellipsis and gapping would show the same movement effects as sentences without gapping. In the next section, I will show that the latter option is correct.

2.2 Q-er movement in sentences with gapping

This section shows that there is evidence for Q-ER movement in sentences where ER is elided as the result of gapping.

First, sentences with gapping show the same argument-adjunct distinction regarding Q-ER pronominalisation as sentences without gapping. That is, NP ellipsis can take place inside arguments but not inside adjuncts. The examples in (8), repeated from (2), show that NP ellipsis can target the argument *vijf boeken* in sentences with and without gapping. In contrast, (9) involves NP ellipsis inside an adjunct. The baseline sentence (9-a), without ellipsis, is grammatical. (9-b) shows that ER pronominalisation

cannot take place in a sentence without gapping; (9-c) shows that the ungrammaticality is retained when there is gapping. We can conclude that, even though ER is not realised in sentences with gapping, it still undergoes movement, leading to the observed difference between arguments and adjuncts.

- (8) a. Ik heb twee boeken gelezen, en jij hebt er vijf geschreven.
 I have two books reads and you have ER five written
 b. Ik heb twee boeken gelezen, en jij ~~hebt er~~ vijf geschreven.
 I have two books read and you have ER five written
 ‘I have read two books, and you have written five books.’
- (9) a. Joost is vijf keer gekust, en Anne is twee keer omhelsd.
 Joost is five times kissed and Anne is two times hugged
 b. *Joost is vijf keer gekust, en Anne is er twee omhelsd.
 Joost is five times kissed and Anne is ER two hugged
 c. *Joost is vijf keer gekust, en Anne ~~is er~~ twee omhelsd.
 Joost is five times kissed and Anne is ER two hugged
 ‘Joost was kissed five times, and Anne was hugged two times.’

Parasitic gaps are also licensed in sentences with gapping and NP ellipsis. (10-a) is another example of parasitic gap licensing by Q-ER movement in a sentence without gapping. (10-b) demonstrates that, even if ER is elided as the result of gapping, the parasitic gap can occur.

- (10) a. Ik heb twee boeken verkocht, en jij hebt er [zonder *pg* te lezen]
 I have two books sold and you have ER without to read
 drie weggedaan.
 three got.rid.of
 b. Ik heb twee boeken verkocht, en jij ~~hebt er~~ [zonder *pg* te lezen]
 I have two books sold and you have ER without to read
 drie weggedaan.
 three got.rid.of
 ‘I have sold two books, and you have gotten rid of three books without reading them.’

Finally, Q-ER pronominalisation of the noun of the first conjunct of a coordination in a sentence with gapping leads to ungrammaticality due to a violation of the Coordinate Structure Constraint. Example (11-a) demonstrates the ungrammaticality of Q-ER pronominalisation of a single conjunct in a sentence without gapping. (11-b), in which gapping and ER deletion have taken place, is equally ungrammatical.

- (11) a. *Ik heb gisteren twee boeken verkocht, en jij hebt er vandaag
 I have yesterday two books sold and you have ER today
 drie en twee tijdschriften gekregen.
 three and two magazines received

- b. *Ik heb gisteren twee boeken verkocht, en jij ~~hebt er~~ vandaag
 I have yesterday two books sold, and you have ER today
 drie en twee tijdschriften gekregen.
 three and two magazines received
 ‘I have sold two books yesterday, and you have received three books
 and two magazines today.’

In conclusion, sentences with and without gapping behave in parallel concerning the effect of Q-ER movement; even though ER is absent in sentences with gapping, underlyingly it must be present and move. The underlying structure of a sentence with gapping and ER is then as in (12). The (linearly) first instance of ER is elided by gapping, the second instance of ER is the movement trace.

- (12) Ik heb twee boeken gelezen, en jij ~~hebt er~~ vijf ~~er~~ geschreven.
 I have two books read and you have ER five ER written
 ‘I have read two books, and you have written five.’

2.3 Gapping and Q-er deletion

The previous section established that Q-ER moves in sentences with gapping. This means that gapping deletes Q-ER from its NP-external position. This section provides an analysis of this observation.

I adopt the analysis of gapping in Dutch by Broekhuis (2018); Broekhuis and Corver (2019). Broekhuis; Broekhuis and Corver observe that there are two types of remnants of gapping in Dutch. The first type are phrases that can undergo A-bar movement in non-reduced clauses (see also Neijt (1979)). For example, the remnants of gapping in (13-a) (in small caps) can be questioned (13-b) or topicalised (13-c) in the non-reduced version of the sentence. These types of remnants always have a correlate in the antecedent clause and make up the majority of gapping remnants.

- (13) a. Ik lees boeken en JIJ ~~leest~~ TIJDSCHRIFTEN
 I read books and you read magazines
 ‘I am reading books and you are reading magazines.’
 b. WIE leest tijdschriften?
 who reads magazines
 ‘Who is reading magazines?’
 c. TIJDSCHRIFTEN lees jij graag.
 magazines read you happily
 ‘Magazines, you like to read.’

The second type of gapping remnants are focus particles like *ook* ‘also’ and polarity particles like *niet* ‘not’. These remnants do not necessarily need a correlate in the preceding clause, as illustrated in (14): *ook* is new the clause with gapping. Interestingly, even if there is a correlate in the preceding clause for the negative particle *niet*, *niet* must survive gapping; it cannot be elided (15).

- (14) Jan heeft Els gezien, en PETER ~~heeft~~ MARIE ook gezien.
 Jan has Els seen and Peter has Marie also seen

‘Jan saw Els, and Peter also saw Marie.’

- (15) a. Jan heeft Els niet gezien, en PETER ~~heeft~~ MARIE niet ~~gezien~~
 Jan has Els not seen and Peter has Marie not seen
 b. *Jan heeft Els niet gezien, en PETER ~~heeft~~ MARIE ~~niet~~ gezien
 Jan has Els not seen and Peter has Marie not seen
 ‘Jan has not seen Els, and Peter has not seen Marie.’ (Broekhuis, 2018, 30)

Broekhuis (2018); Broekhuis and Corver (2019) argue that these two types of remnants have in common that they are in A-bar specifiers, i.e. the specifiers of CP and clause-medial topic, focus, and negation phrases. Remnants of the first type (phrasal remnants) move to these specifiers, whereas remnants of the second type (focus and polarity particles) are externally Merged in these specifiers. Broekhuis (2018); Broekhuis and Corver (2019) propose that gapping elides everything but A-bar specifiers. This explains why only elements that can move to A-bar specifiers can be remnants of gapping: only they have the means to survive gapping by A-bar movement; and why elements like *niet* must survive gapping: these elements are always in A-bar specifiers and can therefore never be elided via gapping.

Returning to Q-ER deletion, the question is now what kind of position Q-ER is in after movement from the NP. If Q-ER is in an A-bar specifier, the analysis of gapping predicts that Q-ER survives gapping. If Q-ER is not in an A-bar specifier, the prediction is that Q-ER elides under gapping. I will now show that the latter is the case: Q-ER is not in an A-bar specifier in its derived position.

The first argument that Q-ER is not in an A-bar specifier is that Q-ER cannot be in Spec,CP. This is illustrated in (16), where the sentence-initial position must be Spec,CP since the finite auxiliary is in C here (preceding the subject in Spec,TP).

- (16) *Er heeft Anne twee gelezen.
 ER has Anne two read
 intended: ‘Anne has read two.’

A likely explanation for the ungrammaticality of (16) is that ER is a weak pronoun, and weak pronouns may not occur in Spec,CP (Zwart, 1997). Strong pronouns must be used instead. The strong version of ER is *daar* ‘there’, illustrated in (17) with the prepositional function. However, *daar* cannot be used as a quantitative pronoun, neither in the sentence-medial (18-a) nor in the sentence-initial position (18-b).

- (17) a. Anne zit er op.
 Anne sit ER on
 b. Anne zit daar op
 Anne sit there on
 ‘Anne is sitting on it.’
 (18) a. *Anne leest daar twee.
 Anne reads there two
 b. *Daar leest Anne twee.
 there reads Anne two

intended: ‘Anne is reading two.’

The same facts hold for the *wh*-version of ER, *waar* ‘where’. *Waar* can be used to question the complement of a preposition (19-a) but not the noun pronominalised by ER (19-b).

- (19) a. Waar zit Anne op?
 where sits Anne on
 ‘What is Anne sitting on?’
 b. *Waar leest Anne twee?
 where reads Anne two
 intended: ‘What is Anne reading two of?’

These facts demonstrate that the quantitative pronoun ER may never be in Spec,CP, most likely because there is no strong pronoun with the quantitative function.

The other specifiers that Broekhuis (2018); Broekhuis and Corver (2019) identify as landing sites for A-bar movement are specifiers of clause-medial phrases. It is possible that these specifiers too are accessible to strong elements only, but this is hard to test due to clause-medial scrambling in Dutch. However, it can be demonstrated otherwise that Q-ER is not in a clause-medial A-bar specifier. If it were, then we would expect it to be able to undergo successive cyclic movement from that position to a similar clause-medial position in a higher clause. This is not possible: Q-ER is clause-bound, see (20). This is the second argument that Q-ER is not in an A-bar specifier.

- (20) a. Jan vertelde mij [dat hij er één meegenomen had].
 Jan told me that he ER one with-taken had
 b. *Jan vertelde er mij [dat hij er één meegenomen had].
 Jan told ER me that he ER one with-taken had
 ‘Jan told me that he had taken one of them with him.’ (Broekhuis & den Dikken, 2012, 937)

I conclude that Q-ER is not in an A-bar specifier in its derived position. It now follows straightforwardly that Q-ER is deleted under gapping in Broekhuis (2018); Broekhuis and Corver (2019)’s analysis of gapping in Dutch. Since gapping elides everything but A-bar specifiers, Q-ER must undergo deletion as the result of gapping. This accounts for the data in (21), repeated from (2).

- (21) a. Ik heb twee boeken gelezen, en jij hebt *(er) vijf geschreven.
 I have two books read and you have ER five written
 b. Ik heb twee boeken gelezen, en jij ~~hebt~~ *(er) vijf geschreven.
 I have two books read and you have ER five written
 ‘I have read two books, and you have written five.’

2.4 Intermezzo: landing sites

If Q-ER does not move to an A-bar specifier, as concluded in the previous section, then we are faced with the question to what kind of position Q-ER does move. We can reject the idea that Q-ER placement is phonological: the evidence for Q-ER movement

clearly shows that Q-ER moves in syntax. This leaves us with two options: Q-ER moves to an A-position, or Q-ER moves to a head position.

The first option, that Q-ER moves to an A-position, can be rejected. In contrast to expletive ER in for example impersonal passives, Q-ER cannot be in the subject position Spec,TP. The contrast is illustrated in (22).²

- (22) a. Er wordt hard gewerkt.
 ER is hard worked
 ‘Hard work is going on.’
 b. *Er staan twee in de tuin.
 ER stand two in the garden
 intended ‘There are two in the garden.’

The only remaining option is that Q-ER moves to a head position. This would imply that Q-ER is a head. In the next section, I review the arguments from previous literature on the syntactic status of Q-ER. I will conclude that, when looking at the data carefully, we should indeed conclude that Q-ER is a head.

3 The syntactic status of Q-er

Previous work on Q-ER has shown that Q-ER can cooccur with certain nominal modifiers, but not others, in Netherlandic (NL-)Standard Dutch. Specifically, Q-ER can be modified by restrictive relative clauses and PP adjuncts, but not by PP complements and adjectives (Barbiers, 2017; Blom, 1975-1976; Corver et al., 2009; Kranendonk, 2010).³ This is illustrated in the following examples:

- (23) Restrictive relative clause: ✓
 Over broeken gesproken, ik heb er twee die ik zelf gemaakt heb
 about trousers spoken I have ER two that I self made have
 verkocht.
 sold
 ‘Speaking about trousers, I have sold two that I made myself.’
 (24) PP adjunct: ✓
 Over collega’s gesproken, ik heb er twee uit Spanje uitgenodigd.
 about colleagues spoken I have ER two from Spain invited
 ‘Speaking about colleagues, I have invited two from Spain.’
 (25) PP complement: *
 *Over beloftes gesproken, ik heb er twee aan Marie geschonden.
 about promises spoken, I have ER two to Mary violated

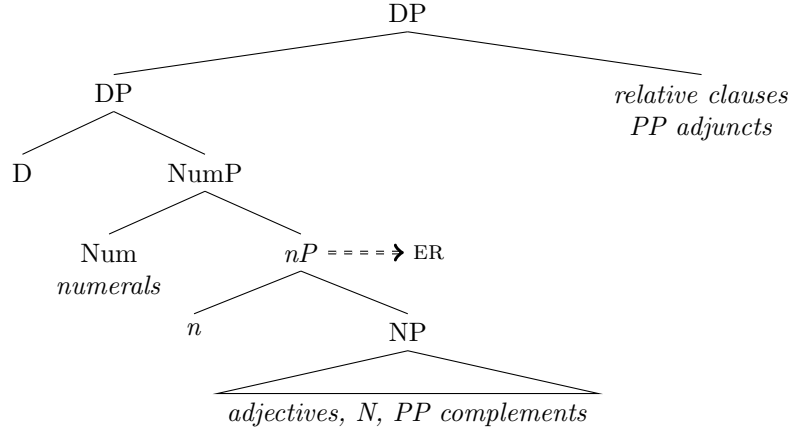
²In the ungrammatical (22)b, ER presumably has a dual expletive and quantitative function. This is generally allowed, see Bennis (1986); the example shows that the quantitative use is not available in Spec,TP.

³Q-ER can cooccur with adjectives in Flemish Dutch. I will address this in section 3.1.

- (26) Adjective: *
- Over broeken gesproken, ik heb (*er) twee groene verkocht.
 about trousers spoken, I have ER two green sold
- ‘Speaking about promises, I have broken two to Mary.’
 ‘Speaking about trousers, I sold two green ones.’

Based on these observations, [Barbiers \(2017\)](#); [Kranendonk \(2010\)](#) conclude that the domain that Q-ER pronominalises contains PP complements and adjectives, but not restrictive relative clauses and PP adjuncts. For example, [Kranendonk \(2010\)](#) proposes that Q-ER pronominalises *nP*. Adjectives and PP complements are introduced below *nP*, so they are included in the pronominalisation and cannot cooccur with Q-ER; relative clauses and PP adjuncts, however, are introduced as high adjuncts to DP, so they are outside of the domain of pronominalisation, and can co-occur with Q-ER. The corresponding structure to this analysis is given in (27).

- (27) (cf. [Kranendonk \(2010\)](#))



If Q-ER is a head, this analysis cannot be right: the domain of a head cannot include modifiers. In the next two sections, I argue that adjectives and PP complements should not be considered part of the domain of Q-ER; instead, the inability of these modifiers to cooccur with Q-ER is due to other factors: competition between Q-ER and the licenser of adjectival remnants of NP ellipsis, and a general ban on complements in NP ellipsis contexts.

3.1 On the cooccurrence of adjectival remnants of NPE and Q-er

As it stands, the structure in (27) predicts that Q-ER and adjectives are in complementary distribution, because Q-ER pronominalisation always includes adjectives. While this prediction is borne out in NL-Standard Dutch, it is not in many dialects. For example, in Schellebelle Dutch (East-Flanders), Q-ER and adjectives do cooccur:

In NP ellipsis contexts, adjectives exhibit the same inflectional suffixes, illustrated in (32).

- (32) a. Speaking about cars(M)... hij het pas ne nij-en gekocht.
 he has recently a new-M bought
 ‘He has recently bought a new one.’
 b. Speaking about cows(F)... hij het pas een bont-e gekocht.
 he has recently a colourful-F bought
 ‘He has recently bought a colourful one.’
 c. Speaking about rabbits(N)... hij het pas een zwart-Ø gekocht.
 he has recently a black-N bought
 ‘He has recently bought a black one.’ (Corver & van Koppen, 2011,
 408)

Corver and van Koppen (2011) argue that these two types of dialects make use of different licensing strategies for NPE with adjectival remnants. Dialects like NL-Standard Dutch use a pronominalisation strategy, i.e., the noun is not elided but pronominalised (much like English *one*-insertion in NP ellipsis contexts). The pronominaliser in these varieties is the bound morpheme *-e*. The pronominaliser does not express gender, resulting in the absence of a gender contrast in NPE contexts. Dialects like Asten Dutch, on the other hand, have real NP ellipsis, i.e. the noun is not pronominalised but empty. The empty noun is licensed via the inflection on the adjective.

I will now show that there is a correlation between the strategy that a dialect uses to license NPE with adjectival remnants, and the ability of Q-ER to cooccur with adjectives: Q-ER cooccurs with adjectival remnants of NPE in dialects that use the ellipsis strategy only.⁵ This follows if Q-ER is a head: in dialects that use the ellipsis strategy to license NPE with adjectival remnants, the noun is empty; therefore, Q-ER can be inserted as a pronominalisation of the N head. In dialects that use the pronominalisation strategy to license NPE with adjectival remnants, the noun is pronominalised by the bound morpheme *-e*. This blocks insertion of Q-ER in N.

The data that show this correlation come from the database Diversity in Dutch DP Design (DiDDD) (Corver et al., 2011). As a proxy for NPE licensing strategy, I look at the behaviour of the neuter in NPE: In varieties like NL-Standard Dutch, that use the pronominalisation strategy, a neuter adjective has the same form as a common, or masculine and feminine, adjective in NPE contexts. In varieties like Asten Dutch, that use the ellipsis strategy, a neuter adjective is distinct from the common, or masculine and feminine, adjective in NPE contexts. The following test sentences were used.⁶

⁵Kranendonk (2010) observes a highly similar relation between the ability of Q-ER to cooccur with adjectives, and the presence of gender marking on indefinite articles. This relation exists because the dialects with gender marking on indefinite articles almost fully overlap with the dialects that use the ellipsis strategy for NPE licensing. However, I believe gender marking on indefinite articles is a confound, not an explanation of the distribution of Q-ER across dialects.

⁶ Sentence (33-c) was used in most of the Netherlandic area and (Dutch and Belgian) Limburg. Sentence (33-d) was used in Flanders (except Limburg) and Zeeland-Flanders in the Netherlands. The reason is that both the nouns *boek* 'book' and *konijn* 'rabbit' have different genders across the language area: *boek* is masculine in parts of Flanders, and *konijn* is masculine in Limburgian. Using (33-c) and (33-d) in their respective areas obviates this issue.

- (33) a. Over auto's gesproken, hij heeft pas een nieuwe gekocht. (M)
 about cars spoken he has recently a new bought
 'Speaking about cars, he has recently bought a new one.'
- b. Over koeien gesproken, hij heeft pas een bonte gekocht. (F)
 about cows spoken he has recently a colourful bought
 'Speaking about cows, he has recently bought a colourful one.'
- c. Over boeken gesproken, hij heeft pas een leuke gelezen. (N)
 about books spoken he has recently a nice read
 'Speaking about books, he has recently read a nice one.'
- d. Ik heb een wit konijn en jij een zwart(e). (N)
 I have a white rabbit and you a black
 'I have a white rabbit and you have a black one.'

The results, depicted in figure 1, show that there is a clear distinction between northern and southern varieties of Dutch. In southern varieties, the neuter is distinct from the common, or masculine and feminine, whereas in northern varieties, the neuter is identical to the other gender(s). This means that southern varieties use the ellipsis strategy, whereas northern varieties use the pronominalisation strategy.

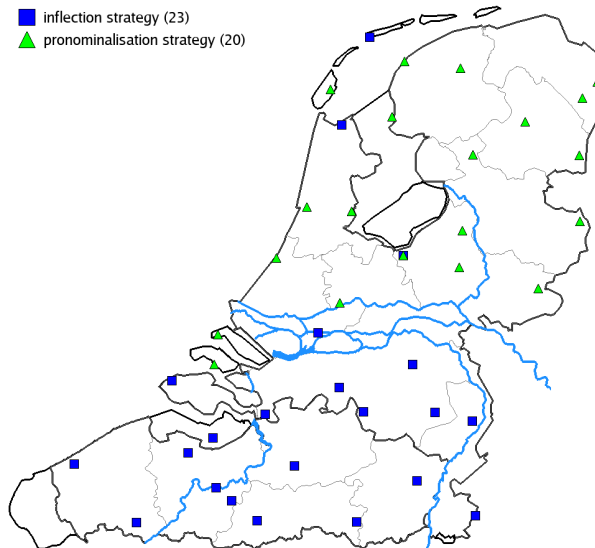


Fig. 1 Licensing strategy for NP ellipsis with adjectival remnants (data: DiDDD)

At the beginning of this section, we saw an example of cooccurrence of Q-ER and adjectives in a Flemish dialect (Schellebelle). If it is correct that Q-ER is a head, we predict that it can be used in all varieties that use the ellipsis strategy for NPE, i.e. all southern varieties. Figure 2 shows where Q-ER can be used in combination with a remnant of NPE that consists of a numeral and an adjective. The figure is based on the sentences in (34), again from the DiDDD (Corver et al., 2011).

- (34) a. Hij heeft vijf rode appels en ik heb vier groene.
 he has five red apples and I have four green
 ‘He has five red apples and I have four green ones.’
 b. Ik heb vier zwarte kippen en hij heeft tien witte.
 I have four black chickens and he has ten white
 ‘I have four black chickens and he has ten white ones.’

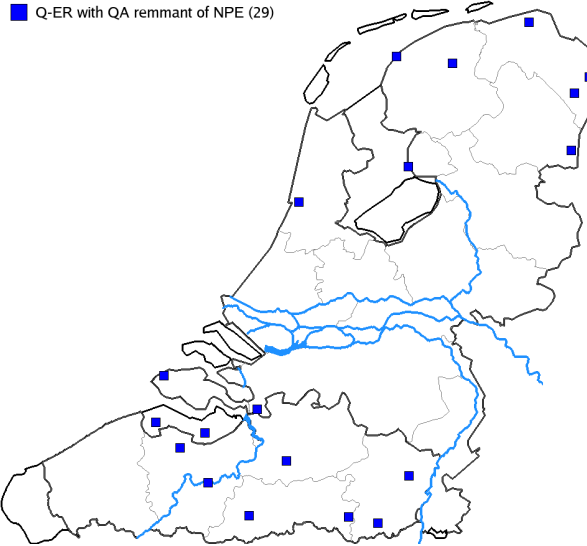


Fig. 2 Cooccurrence of Q-ER with Q+A remnants of NP ellipsis (data: DiDDD)

Two areas on the map behave as expected. First, the Flemish and Zeeland dialects allow for cooccurrence of Q-ER and adjectives. This is well known (de Rooij, 1991; De Schutter, 1992; Kranendonk, 2010; Sleeman, 2024) and expected under the head analysis of Q-ER, because these varieties use the ellipsis strategy for NPE with adjectival remnants, so ER can be used to spell out N. Second, the central Netherlandic area also behaves as expected in not allowing Q-ER with adjectival remnants of NPE, because these varieties use a pronominalisation of N to license adjectival remnants, blocking ER insertion. The other two areas, i.e. the north and south of the Netherlands, do not show the expected pattern at first sight, and deserve more comment.

Focusing first on the northern Dutch and Frisian varieties, the head analysis of Q-ER predicts that Q-ER is blocked with adjectival remnants of NPE, because these varieties use the pronominalisation strategy with adjectives. Yet, figure 2 shows that ER does cooccur with adjectives in this area. However, ER shows different behaviour in these varieties more generally: specifically, ER is optional with remnants of NPE (Kranendonk, 2010). This is unexpected if ER is needed to license NPE. In fact, the consensus in the literature on Frisian is that Frisian does not have quantitative ER at all, and that quantificational remnants of NPE are always bare, as in (35) (E. Hoekstra, 2015; J. Hoekstra, 1991).

- (35) Ik ha (*der) fjouwer opiten.
 I have ER four eaten
 ‘I have eaten four.’ (J. Hoekstra, 1991, 61)

If Frisian does not have Q-ER, then the question is what kind of ER we are dealing with in an example like (36) from the DiDDD. Hoekema (1983) suggests that ER in NPE contexts in Frisian is in fact locative ER; possibly, using ER in such contexts is the result of interference from Dutch, as suggested by E. Hoekstra (2015). In any case, it seems that ER in (36) is not Q-ER, and therefore we do expect it to be in competition with the licenser of adjectival remnants of NPE. I conclude that the Frisian varieties are not a counterexample to the proposed generalisation.⁷

- (36) He hat fûf reade apels en ik ha der fjouwer (griene).
 he has five red apples and I have ER four green
 ‘He has five red apples and I have four (green) ones.’ Bergum Frisian
 (DiDDD)

The second area that deserves comment is the southern Netherlandic area. This region uses the ellipsis strategy for NPE with adjectival remnants, but there is no cooccurrence of adjectival remnants of NPE and Q-ER. This is not necessarily problematic for the proposed generalisation, since it might be one-way: If N is not pronominalised in the context of adjectival remnants of NPE, a variety could use Q-ER to pronominalise N, but it not need do so. However, Hendriks (2025) reports on seven North-Brabant speakers that do allow cooccurrence of Q-ER and remnants of NPE that consist of a numeral and an adjective. It is therefore possible that the southern Netherlandic area does allow combinations of Q-ER and adjectives, but that it was simply not recorded in the DiDDD interviews, because ER is always optional in this context.

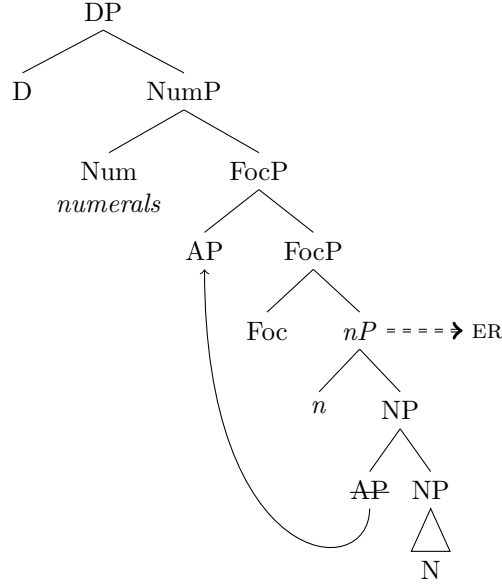
In conclusion, I have demonstrated that there is a correlation between ellipsis licensing strategy with adjectival remnants and cooccurrence of Q-ER and adjectives. Varieties that pronominalise N to license adjectival remnants of NPE do not allow Q-ER in combination with adjectives. Varieties that use ellipsis to license adjectival remnants of NPE do allow Q-ER in combination with adjectives. In other words, Q-ER can be used with adjectival remnants of NPE only in those varieties in which N is not pronominalised in the context of adjectival ellipsis remnants. This pattern follows in a very straightforward manner if Q-ER is also a pronominalisation of N.

3.1.1 Alternative analysis: focus movement

Corver et al. (2009); Kranendonk (2010) propose an alternative analysis of the cooccurrence of Q-ER and adjectives, which is compatible with the idea that Q-ER pronominalises a phrase (*nP*) that includes adjectives. In particular, they propose that adjectives move out of the domain of Q-ER to a NP-internal focus position, and can therefore “escape” Q-ER pronominalisation of the *nP*. This is schematically represented in (37).

⁷By extension, I assume that the same interpretation holds for the northern Dutch varieties due to their geographical and linguistic proximity to Frisian, but further data collection would be needed to confirm this.

(37)



The motivation for this proposal mainly comes from Frisian. In Frisian, indefinite adjectival remnants of NPE can take the suffix *-(e)n* (38) (as well as *-e* or *-enien*, cf. Barbiers (2005); Corver and van Koppen (2009, 2010)). If *-(e)n* is used as an adjectival suffix in non-ellipsis contexts, it expresses emphasis, illustrated in (39) (Tiersma, 1985).

- (38) in saai-en
 a boring-en
 ‘A boring one.’ (Corver & van Koppen, 2009, 6)

- (39) Hy is in dreg-en baas
 he is a tough-en guy
 ‘He is a tough guy.’ (Tiersma, 1985, 46)

Because of this effect on interpretation, Corver et al. (2009); Kranendonk (2010) propose that *-(e)n* is the spell out of a focus head that projects an NP-internal focus phrase. The adjective subsequently moves to the specifier of the FocP to combine with the suffix. Kranendonk (2010) extends this analysis to the southern dialects, although in these dialects there might not be an overt realisation of the focus head.

This analysis predicts that focus movement of the adjective is a prerequisite for the cooccurrence of adjectives and ER; therefore, ER should be allowed when there is a morphological reflex of focus movement, i.e. *-(e)n* on the adjective, but not in other contexts, i.e. when the adjectival suffix is *-e* (I assume that *enien* is a compound suffix consisting of *-(e)n* and *ien* ‘one’, cf. Barbiers (2005); Corver and van Koppen (2011)). This prediction is not borne out. ER cooccurs with adjectives suffixed with *-e* and *-en*, as demonstrated below.⁸

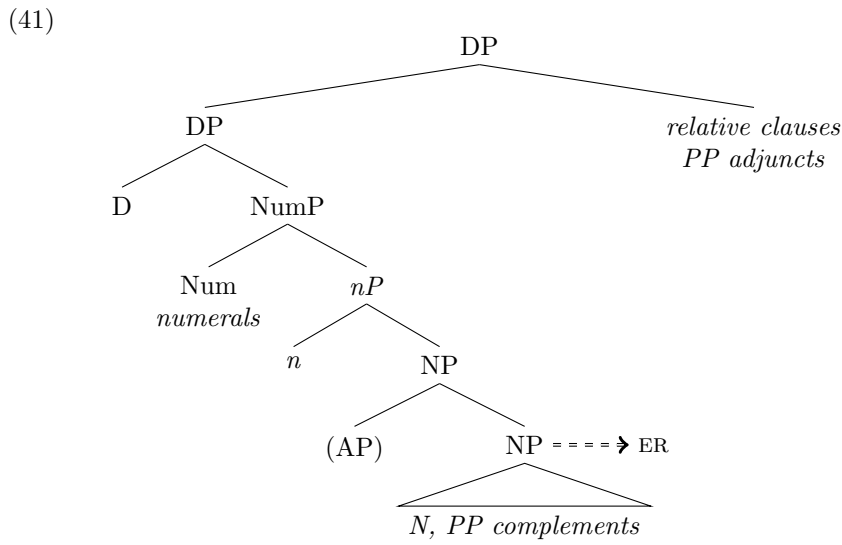
⁸Klazienaveen Dutch is not a variety of Frisian, but a Low Saxon dialect; there are no Frisian examples in the DiDDD in which ER and the adjectival suffix *-(e)n* cooccur.

- (40) a. He hat fûf reade apels en ik ha der fjouwer grien-e.
 he has five red apples and I have ER four green-e
 ‘He has five red apples and I have four green ones.’ Bergum Frisian
- b. ... en ik heb der vier grûn-n.
 and I have ER four green-n
 ‘.. and I have four green ones.’ Klazienaveen Dutch (DiDDD)

In conclusion, the connection between focus movement of the adjective and ER is not supported in morphosyntax. For this reason, I believe that the analysis of the cooccurrence between Q-ER and adjectives based on focus movement can be rejected.⁹

3.2 NP ellipsis and complementation

The previous section established that the domain of Q-ER does not include adjectival modifiers of the noun. Based on this conclusion, the structure for Q-ER can be revised as follows:



⁹Corver et al. (2009); Kranendonk (2010) also discuss examples like (i) from Standard Dutch, in which ER cooccurs with a comparative adjective and the focus particle *nog* ‘even’. Kranendonk suggests that examples of this kind also involve focus movement of the adjective.

- (i) Hij heeft mooie appels, maar ik heb (er) nog mooiere dan hij.
 he has nice apples but I have ER even nicer than he
 ‘He has nice apples, but I have even nicer ones than him.’ (Corver et al., 2009)

While I cannot offer a full analysis of this observation, it seems relevant that other focus particles do not combine with ER. For example, (ii), with the focus particle *al*, is ungrammatical. In other words, there might be something about *nog* that makes ER insertion possible, but I leave a further exploration of this possibility for future work.

- (ii) *Hij heeft veel appels, maar ik heb er al grotere dan hij.
 he has many apples, but I have ER already bigger than he
 ≈ ‘He has many apples, but my ones are already bigger.’

However, if Q-ER is a head, PP complements cannot be part of the domain of Q-ER either. I propose that the ban on modification of Q-ER with PP complements results from a general inability of pronominalised nouns to select arguments. The motivation for this proposal comes from the observation that PP complements are ungrammatical in NPE contexts in general. Consider the following examples of different types of modifiers in Dutch, and their English translations:

- (42) Restrictive relative clause: ✓

Over broeken gesproken, ik heb een groene die ik zelf gemaakt heb
 about trousers spoken I have a green that I self made have
 verkocht.
 sold

‘Speaking about trousers, I have sold a green one that I made myself.’

- (43) PP adjunct: ✓

Over collega’s gesproken, ik heb die nieuwe uit Spanje uitgenodigd.
 about colleagues spoken I have the new from Spain invited

‘Speaking about colleagues, I have invited the new one from Spain.’

- (44) PP complement: *

*Over beloftes gesproken, ik heb een plechtige aan Marie geschonden.
 about promises spoken, I have a solemn to Mary violated

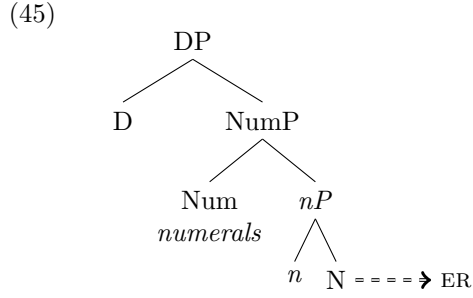
‘*Speaking about promises, I have broken a solemn one to Mary.’

As with Q-ER, adjectival remnants of NPE in Dutch and English can be modified with relative clauses and PP adjuncts, but not with PP complements. As discussed in the previous section, [Corver and van Koppen \(2011\)](#) argue that in Dutch adjectival remnants of NPE, the head noun is pronominalised by the schwa suffix on the adjective. For English, [Panagiotidis \(2003\)](#) argues that *one* is also a N head. Due to their nature as heads, PP complements cannot be part of the domain that is pronominalised; yet, PPs are excluded as complements to pronominalised nouns. This means that PP complements must be excluded on independent grounds. Following [Panagiotidis \(2003\)](#), I propose that pro-nouns like English *one*, Dutch adjectival *-e*, and Q-ER cannot occur with PP complements because they do not select arguments; pro-nouns are semantically and syntactically underspecified and do not have theta-roles to assign. As a result, arguments remain unlicensed and their presence results in ungrammaticality.

3.3 Interim summary

This section argued against the claim that the domain of Q-ER pronominalisation includes adjectives and PP complements. I demonstrated that the absence of adjectives (in NL-Standard Dutch) and PP complements in the context of Q-ER is the result of independent properties, specifically NP ellipsis licensing strategy, and the inability

to select arguments. We can conclude that Q-ER is not a phrase but a N head. The structure of a DP with Q-ER pronominalisation is given in (45).



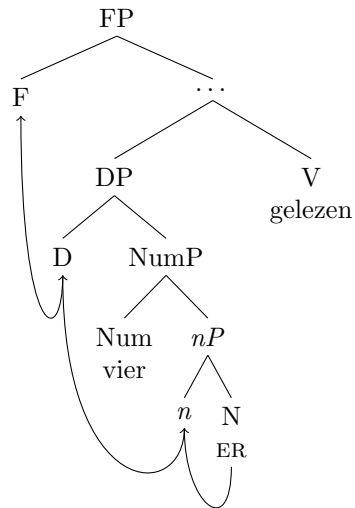
4 Long head movement

The conclusion that Q-ER is a head means that Q-ER movement must be head movement. However, Q-ER movement has properties that are not reminiscent of head movement.

The first property is that Q-ER movement can skip heads, i.e. it violates the Head Movement Constraint (Travis, 1984). The position of Q-ER in a clause is to the right of the finite verb (subject-initial main clause) or subject (non-subject initial main clause or embedded clause) and, if present, indirect object. I am going to identify the landing site for Q-ER as being between TP and *v*P, in a functional projection called FP (cf. Broekhuis (2013, 339)’s Axiom I: “ER must be moved into a unique [+R]-position in the middle field of the clause”). If Q-ER moves from a direct object, it needs to skip (at least) V to get to F. This is illustrated in (46).¹⁰

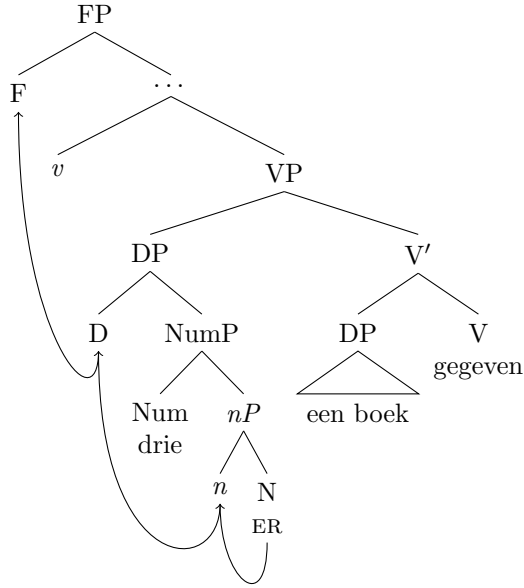
- (46) Ik heb er vier gelezen.
 I have ER four read
 ‘I have read four ones.’

¹⁰Depending on the analysis of numerals as heads or phrases, Q-ER may also need to skip Num. The tree structures in this paper treat numerals as Num heads, but this is for presentational purposes and not as an argument that ER needs to skip heads.



A different issue arises when Q-ER is part of an indirect object or a subject, because in this scenario Q-ER is not extracted from the complement of a head, but from a phrase embedded in the complement. In this situation, we expect only phrasal movement to be possible, cf. [Pesetsky and Torrego \(2001\)](#)'s Head Movement Generalisation. Head movement should be excluded but is what we find here, as illustrated in (47) with movement from an indirect object.

- (47) Ik heb er drie een boek gegeven.
 I have ER three a book given
 'I have given three (people) a book.'



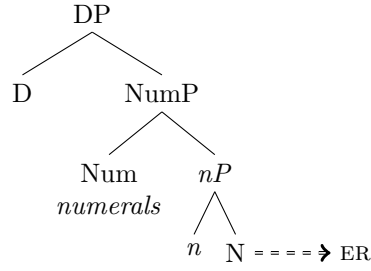
A final property is that Q-ER movement licenses parasitic gaps, cf. (48) repeated from section 2.1. However, since Engdahl (1983) it is well-known that only A-bar movement licenses parasitic gaps, so this is not expected if Q-ER movement is head movement.

- (48) Ik heb je er [zonder *pg* in te kijken] één toegestuurd.
 I have you ER without in to look one sent
 ‘I have sent you the book/one without looking in it.’ (adapted from Barbiers, 2017, 168)

In summary, we are faced with the following situation: Q-ER has the structural size of a head, and its landing site is a head. On the other hand, it can move long-distance and license parasitic gaps.

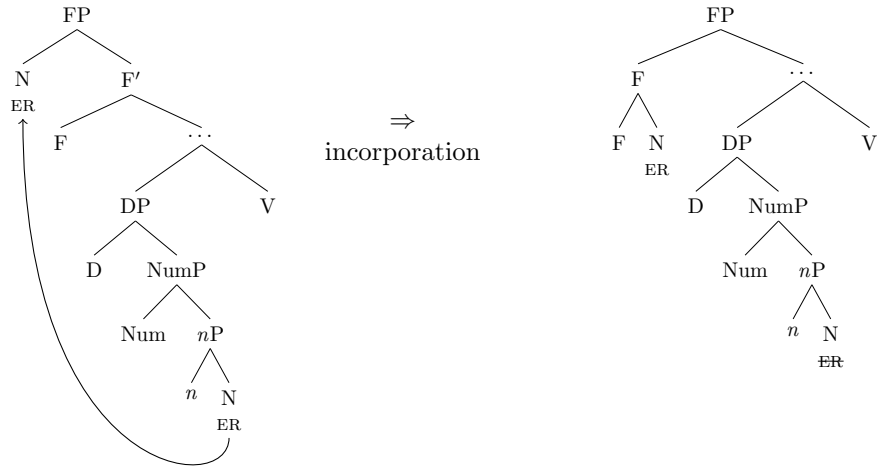
Consider again the structure of Q-ER in (49), repeated from (45). The head **N** is a bare head; it does not have arguments or adjuncts. This means that in bare phrase structure, **N** is in fact ambiguous between head (minimal projection) and phrase (maximal projection). Because of the obligatory absence of adjuncts (adjectives) and arguments (PP complements), **N** always has exactly this structure and is always both minimal and maximal.

(49)



I propose that the ambiguity between head and phrase explains the mixed behaviour of Q-ER. Specifically, I propose that Q-ER moves as a phrase to Spec,FP, and subsequently incorporates into F to form a complex head (cf. [Matushansky, 2006](#)). This is schematically represented in (50). This kind of long head movement has also been proposed for Romance clitics, see [Matushansky \(2006\)](#); [Roberts \(2010\)](#); [Sportiche \(1998\)](#).

(50)



This analysis explains exactly why Q-ER has the properties it has. Q-ER has the size of a N head, which explains why it shows the same behaviour as, and is in competition with, other pro-nouns. However, because Q-ER does not project, it is at the same time a maximal projection and can move as a phrase, i.e. it can undergo long-distance A-bar movement. This allows for parasitic gap licensing, which requires A-bar movement. The head nature of Q-ER forces it to incorporate into the head whose specifier Q-ER lands in. This explains why Q-ER cannot be in an A-bar position and survive gapping, and why it cannot undergo successive cyclic long-distance movement.

A remaining question concerns those varieties of Dutch in which Q-ER can cooccur with adjectives. The analysis of ER requires N to not project; if it does, then it would lose its status as a minimal and maximal projection at the same time. This means that adjectives cannot be adjuncts to NP. I propose that adjectives are adjuncts to

*n*P (restrictive relative clauses and PP adjuncts can also adjoin at this level, contra Barriers (2017); Kranendonk (2010)). If so, N remains bare even in the context of an adjective.

5 Conclusion

This paper has argued that Dutch quantitative ER is a head that undergoes long head movement. Because long head movement is a two-step operation consisting of A-bar movement and head incorporation that can apply to elements that are both minimal and maximal, it was possible to account for the mixed head and A-bar properties of Q-ER. Properties of headedness of Q-ER are: it is in competition with other N heads; it moves to a head position; and movement is also clause-bound. However, Q-ER can move long-distance, and license parasitic gaps, which are properties of A-bar movement. The paper also accounted for the observation that ER obligatorily elides under gapping, as well as for the cooccurrence of Q-ER and adjectives in some Dutch dialects, and the absence of complements in the context of Q-ER pronominalisation.

As was pointed out in the introduction, ER has other functions in addition to the quantitative one, i.e. prepositional, locative, and expletive. It is an open question whether ER is also a head in these other uses, or in other words, whether there is one ER (head) or multiple ERs (head and phrase). One way to go about this is to look at ER interactions (Bech, 1952; Bennis, 1986; Broekhuis, 2013; Huybrechts, 1991). For example, it has been established that one instance of ER may perform multiple functions, but that not all combinations of functions are allowed. Focusing exclusively on the interaction between Q-ER and the other types of ER, the generalisation is that Q-ER may conflate with prepositional and expletive ER, but not with locative ER, as illustrated in (51).

- (51) a. Ik heb er twee over geschreven. (Q + prepositional)
 I have ER two about written
 ‘I have written two about it.’
 b. ... dat er gisteren twee verkocht werden. (Q + expletive)
 that ER yesterday two sold were
 ‘...that yesterday two were sold.’
 c. *Ik heb er twee gekocht. (Q + locative)
 I have ER two bought
 intended interpretation: ‘I have bought two there.’
 actual interpretation: ‘I have bought two.’

If ER conflation requires similar types of ER, this pattern indicates that prepositional ER and expletive ER are heads, but locative ER is a phrase. The full range of ER interactions is complex and requires further investigation, but the head/phrase distinction might be useful to understand the similarities and differences between the different functions of ER.¹¹

¹¹In order to express the intended meaning of (51-c), a strong (phrasal) R-pronoun must be used in addition to Q-ER, as in (i). The fact that this pronoun needs to be strong is most likely an OCP-effect, but it shows that both pronouns need to be syntactically and phonologically present.

Like Dutch, French (as well as other Romance languages) uses a quantitative pronoun *en* in the context of NP ellipsis with indefinite quantificational remnants (52).

- (52) J'en ai lu deux.
 I-EN have read two
 'I have read two.' (Hulk, 1982, 103)

The pronoun *en* has also been argued to be a pronominalisation of N (e.g. Hulk, 1982). The treatment of Q-ER as a head opens up the possibility that Q-*en* and Q-ER can receive identical analyses, i.e. as pronominalisations of N that undergo long head movement to their surface position. If so, we expect *en* to have similar properties as ER when it comes to e.g. behaviour under gapping and parasitic gap licensing. I leave an exploration of these similarities to future research.

Declarations

No competing interests to declare.

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(i) Ik heb er daar twee gekocht.
 I have ER there two bought
 'I have bought two there.'

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